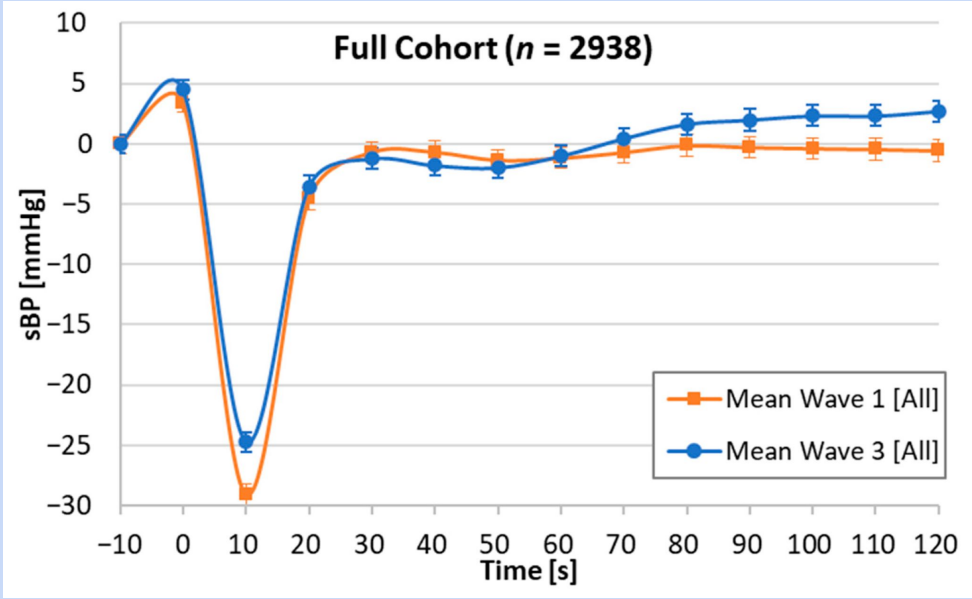
# Antihypotensionator

Sophia White - Ayman Islam - Daniel Dolomount - Zachary Coffey
ECE 8000/10: Final Presentation

# Background, Problem Statement and Objectives

# Background – Orthostatic Hypotension

- Sudden drop in Blood Pressure when a person moves from sitting to standing
- Occurs most commonly in elderly patients and patients living with autonomic nervous system disorders (ex: Parkinson's Disease, Multiple System Atrophy).



# Background - Orthostatic Hypotension

- Results in patients experiencing lightheadedness or fainting, resulting in:
    - Safety concerns
    - Fear of performing daily activities
    - Reduced quality of life
- The underlying causes of Orthostatic Hypotension can not be cured, however the symptoms can be mitigated



# Background - Current Options for Symptom Mitigation

- Lifestyle adaptation
- Compression devices
- Pharmaceutical Options



# Problem Statement

Orthostatic Hypotension patients experience fear of performing their daily activities due to their lightheadedness and fainting. Current symptom mitigation solutions do not adapt to the user’s fluctuating blood pressure, exposing them to risk factors associated with hypertension.

# Objective

- Objective of the Research Group:
    - A method of stabilizing blood pressure in Orthostatic Hypotension patients that adapts to the user’s blood pressure
- Objective for our Capstone Project:
    - Design the necessary hardware and firmware that will:
        - Collect PPG Measurements from a patient's wrist
        - Run Michael's algorithm with PPG data collected from the patient
        - Inflate/deflate a compression device on demand to facilitate the use of Michael's algorithm on Dr. Incognito’s patients

# Primary Success Metrics

1. Device can continuously monitor and provide heart rate data
2. Device can autonomously provide and remove compression
3. Device is fully portable and wearable
4. The device must provide a user-accessible method to recharge or replace the power source
5. Device must be compatible with a mobile application that can collect and display the patient’s blood pressure data

# Design and Alternatives

# Design Options – Heart Rate Data Collection

- Device can continuously monitor and provide heart rate data

**Reflected Green Light PPG:**

✔️ Compatible with Micheal's algorithm

✔️ Easily wearable

✔️ Superior accuracy to Reflected Infrared PPG

✔️ Reliable at temperatures below 15°C

**Reflected Infrared PPG:**

✔️ Compatible with Micheal's algorithm

✔️ Easily wearable

✖️ Inferior accuracy to Reflected Green Light PPG

✖️ Reliable at temperatures below 15°C

**ECG:**

✖️ Not compatible with Micheal's algorithm

✖️ Not easily wearable

✔️ Superior accuracy to Reflected Infrared PPG

✔️ Reliable at temperatures below 15°C

# Design Options - Power Replenishment

- The device must provide a user-accessible method to recharge or replace the power source

**Swappable batteries:**

✖️ Laborious

✔️ Efficient

✖️ Expensive

**Wired charging:**

✔️ Simple

✔️ Efficient

✖️ Flimsy

**Wireless charging**

✔️ Simple

✖️ Less Safe

✖️ Complex & Expensive

# Design Options - App Integration

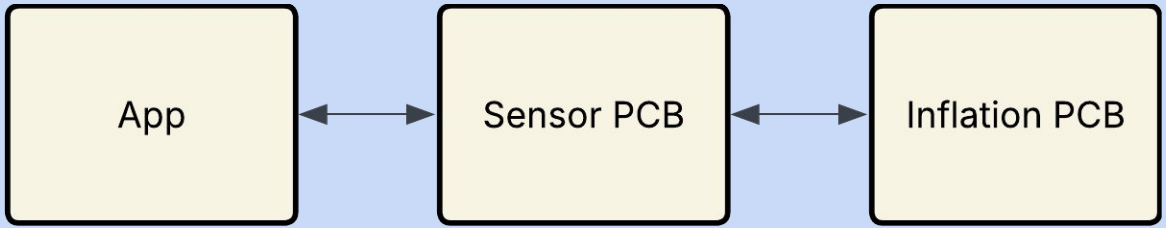
- Device must be compatible with a mobile application

# Final Design - Autonomous Blood Pressure Control

- Decided on a two piece wearable device
- One part will collect data, the other will regulate blood pressure

# Final Design – System Architecture

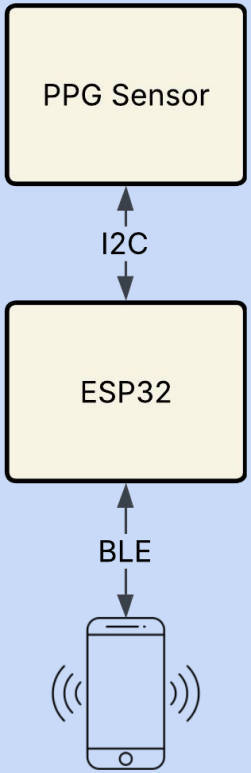
- Three main components:
  - Sensor PCB: Continuously monitors heart rate, sends heart rate data to algorithm
  - Inflation PCB: Autonomously inflates/deflates cuff to regulate blood pressure
  - App: Displays data to users, houses algorithm, communicates with PCBs
- App ↔ Sensor PCB ↔ Inflation PCB: BLE connection
- Sensor PCB ↔ Inflation PCB: WiFi



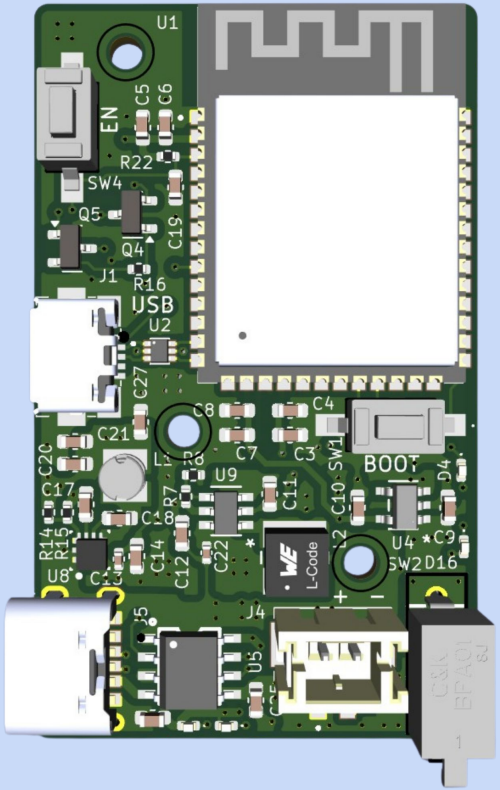
# Sensor Board - Overview

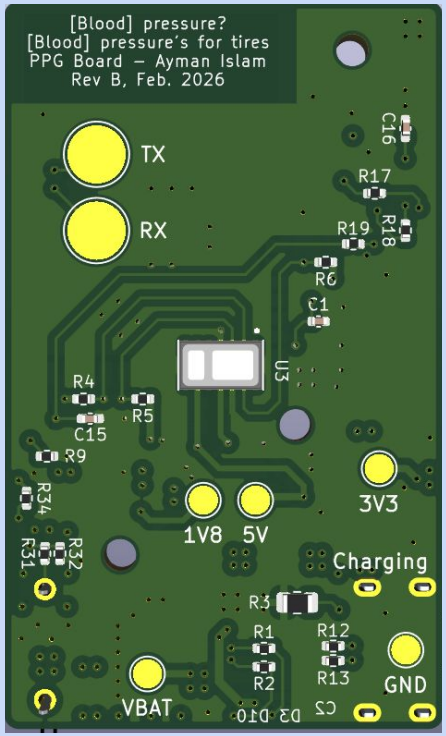
- Custom PCB designed to continuously monitor heart rate
- ESP32 for reading data, and communication
- Uses green LED PPG sensor
- Battery powered & rechargeable





# Sensor Board - Electronics



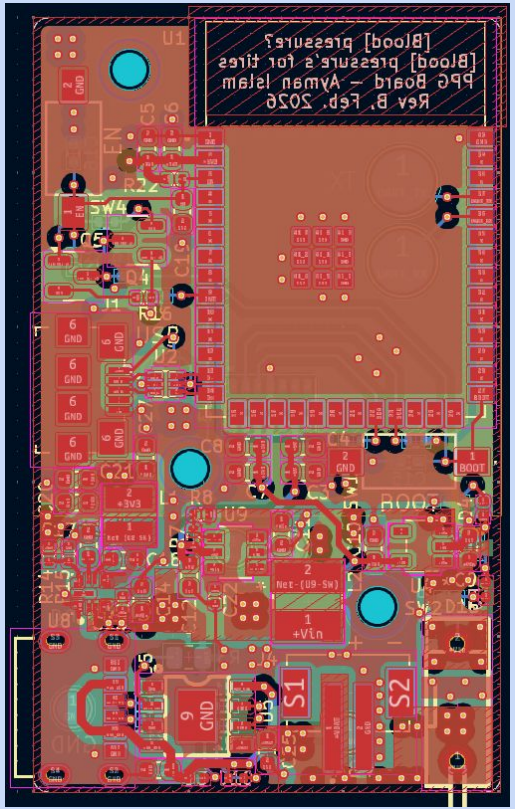
# Sensor Board - Electronics



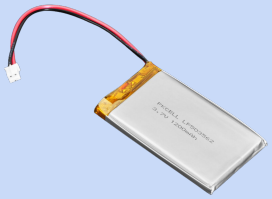



# Sensor Board – Layout

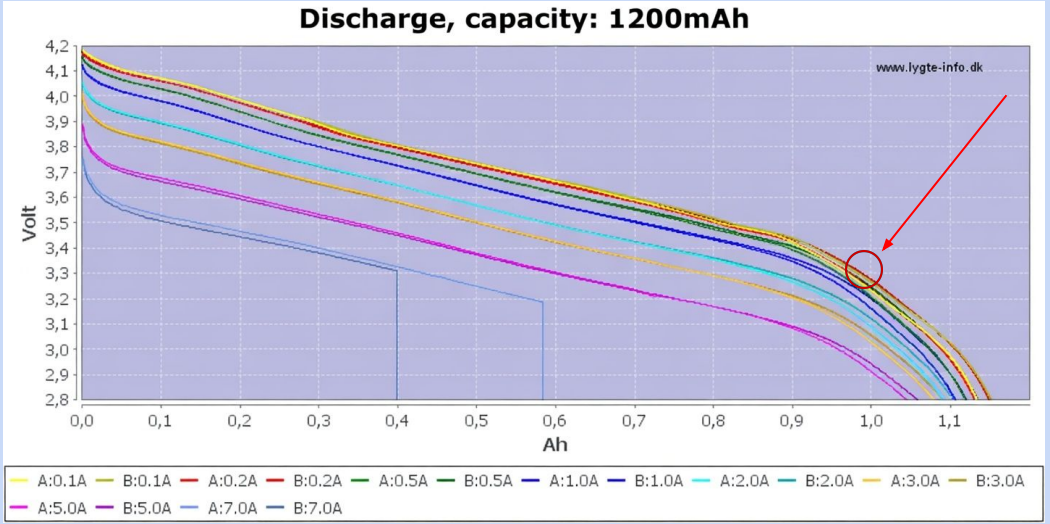
- 5cm x 3cm to match battery size + to fit on wrist
- 4 layer stackup
- Dedicated GND plane for low impedance return paths, signal integrity and thermal management
- Dedicated PWR plane for simple routing and thermal dissipation



# Sensor Board - Battery

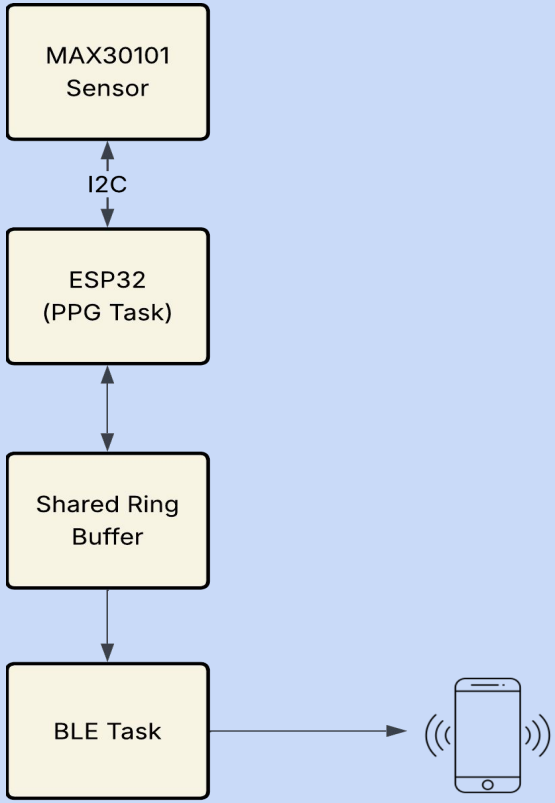


- Standard 3.7V Lithium-ion battery
- Operation range is 4.2V to 3.3V
- 1200mAh, 2A continuous
- Rechargeable, internally protected
- Max operation of board is 200mA
- Curve shows ~1Ah at 3.3V
- 5 hours at max operation
- Charges at 1A, around 1 hour

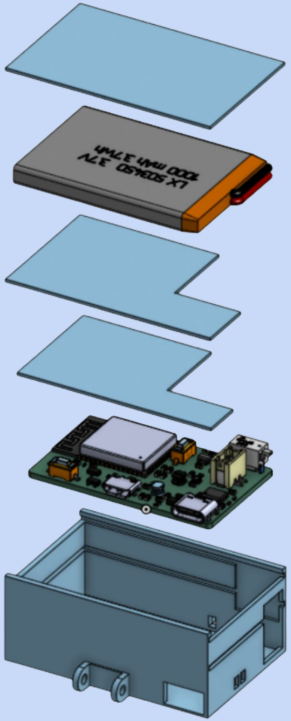


# Sensor Board - Firmware

- Uses ESP-IDF and FreeRTOS
- Firmware written in C
- Concurrent Tasks for:
  - Sensor data acquisition
  - BLE streaming
  - Inter-board communication



# Sensor Board - Mechanical

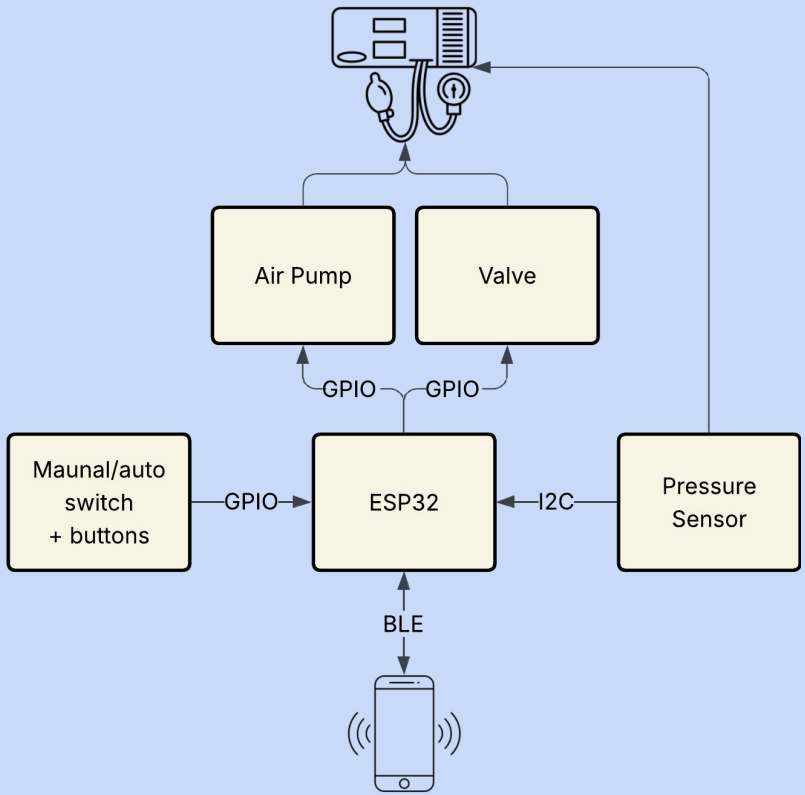




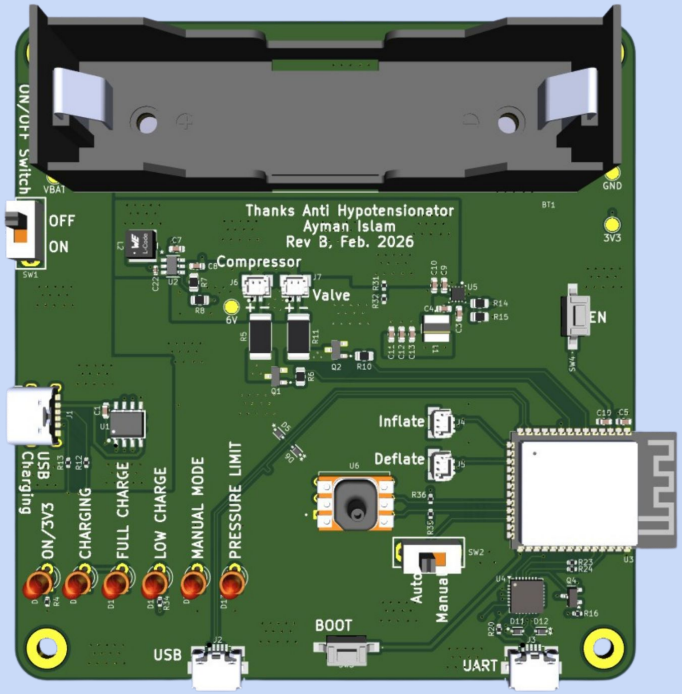


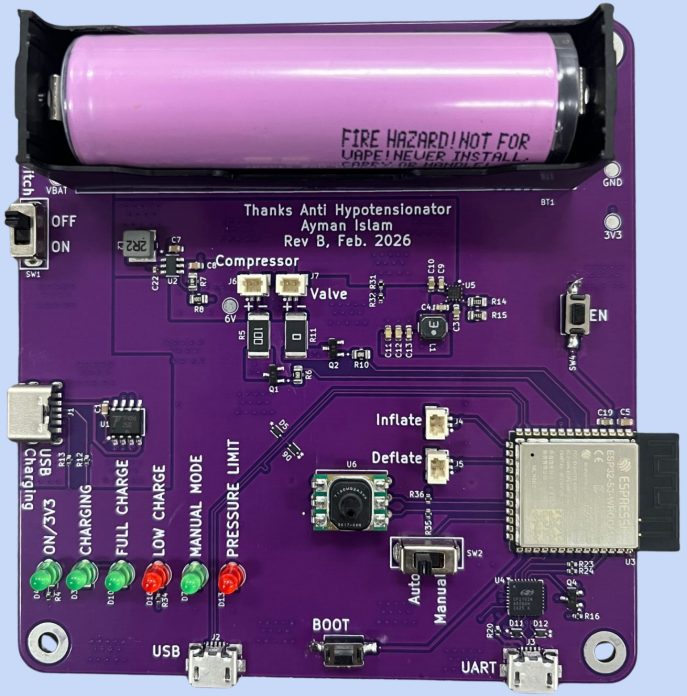
# Inflation Board – Overview

- Custom PCB designed to regulate cuff pressure
- ESP32 for reading data, communication, motor control
- Pressure sensor
- Manual vs auto mode
- Connections for the air pump, valve, and buttons
- Battery powered & rechargeable



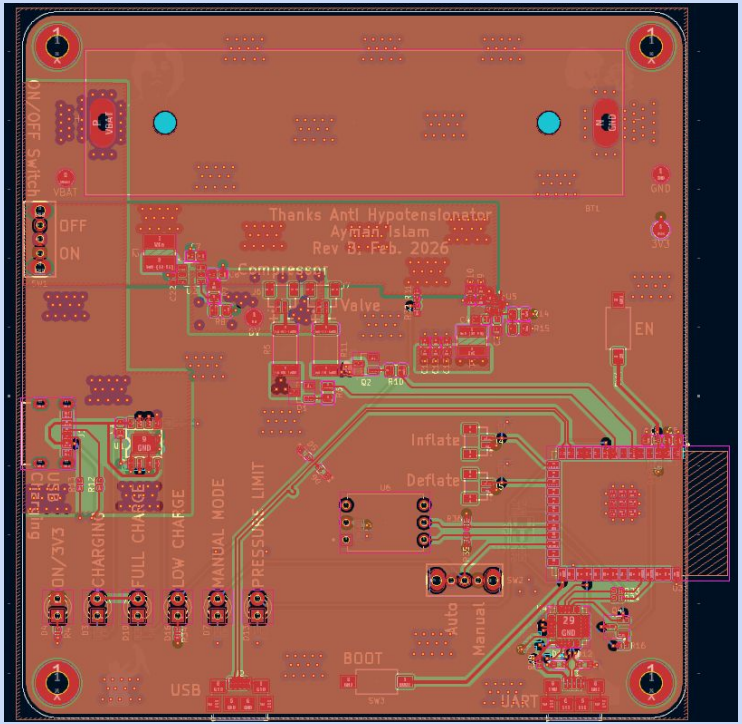
# Inflation Board - Electronics



# Inflation Board – Layout

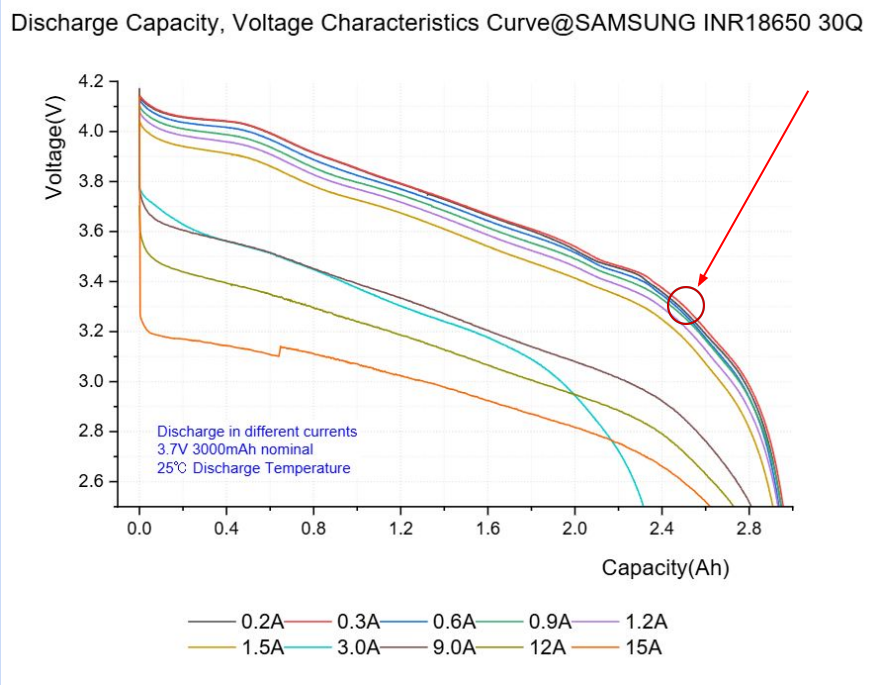
- 10cm x 10cm
- 4 layer stackup
- Dedicated GND plane for low impedance return paths, signal integrity and thermal management
- Dedicated PWR plane for simple routing and thermal dissipation



# Inflation Board - Battery



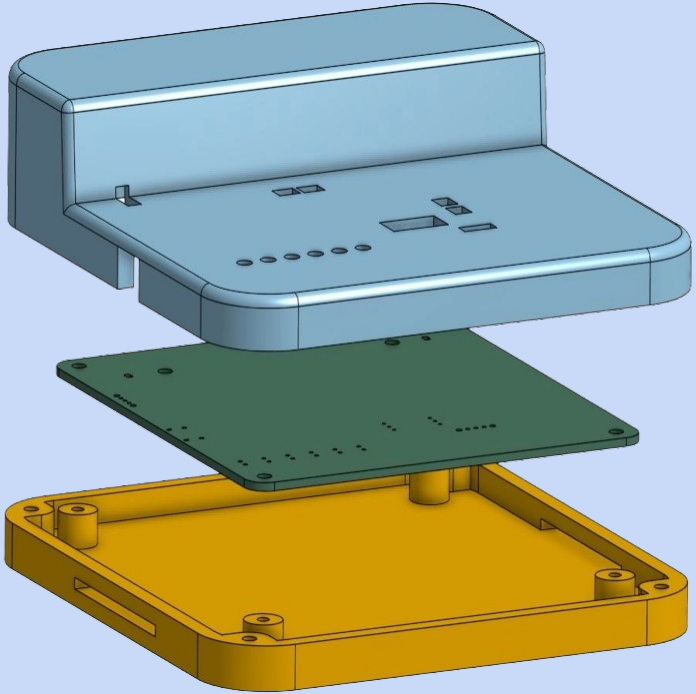
- Samsung 30Q 18650 Lithium-ion
- Operation range is 4.2V to 3.3V
- 3000mAh, 15A continuous
- Rechargeable, internally protected
- Max operation of the board is 600mA
- Curve shows ~2.4Ah at 3.3V
- 4 hours at max operation
- Charges at 1A, around 3 hours

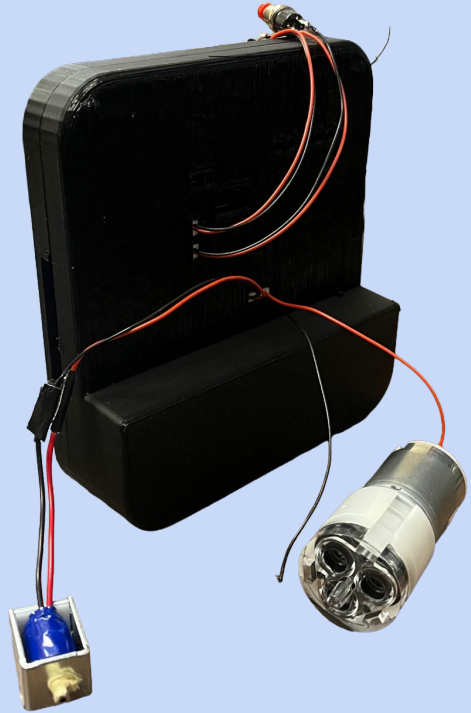


# Inflation Board - Firmware

- Uses ESP-IDF and FreeRTOS
- Concurrent tasks for:
  - Pump control
  - Pressure sensor data acquisition
  - Inter-board communication

# Inflation Board - Mechanical





# APP

# Results and Recommendations

# Final Design Outcomes

✔️ Device can continuously monitor and provide heart rate data

✔️ Device can autonomously provide and remove compression

✔️ Device is fully portable and wearable

✔️ The device must provide a user-accessible method to recharge or replace the power source

✔️ Device must be compatible with a Mobile Application that can collect and display the patient’s blood pressure data

# Limitations

- PPG sensor
    - Off the shelf, not medical grade
- DSP pipeline
- Battery size
    - Limiting size of wrist device

# Risks

- Reliability
  - Extensive patient testing has not been conducted

# Next Steps

- Implement blood pressure algorithm
    - The point of the project
- Make system smaller
    - Smaller microcontroller
    - Smaller batteries
- DSP pipeline improval
- Further develop mechanical design
    - More robust design

# Questions

# References


- https://lygte-info.dk/review/batteries2012/Samsung%20INR18650-30Q%203000mAh%20%28Pink%29%20UK.html
- https://lygte-info.dk/review/batteries2012/Keeppower%2018350%201200mAh%20P1835C2%20%28Black%29%202017%20UK.html
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